

Applied Programming For AI

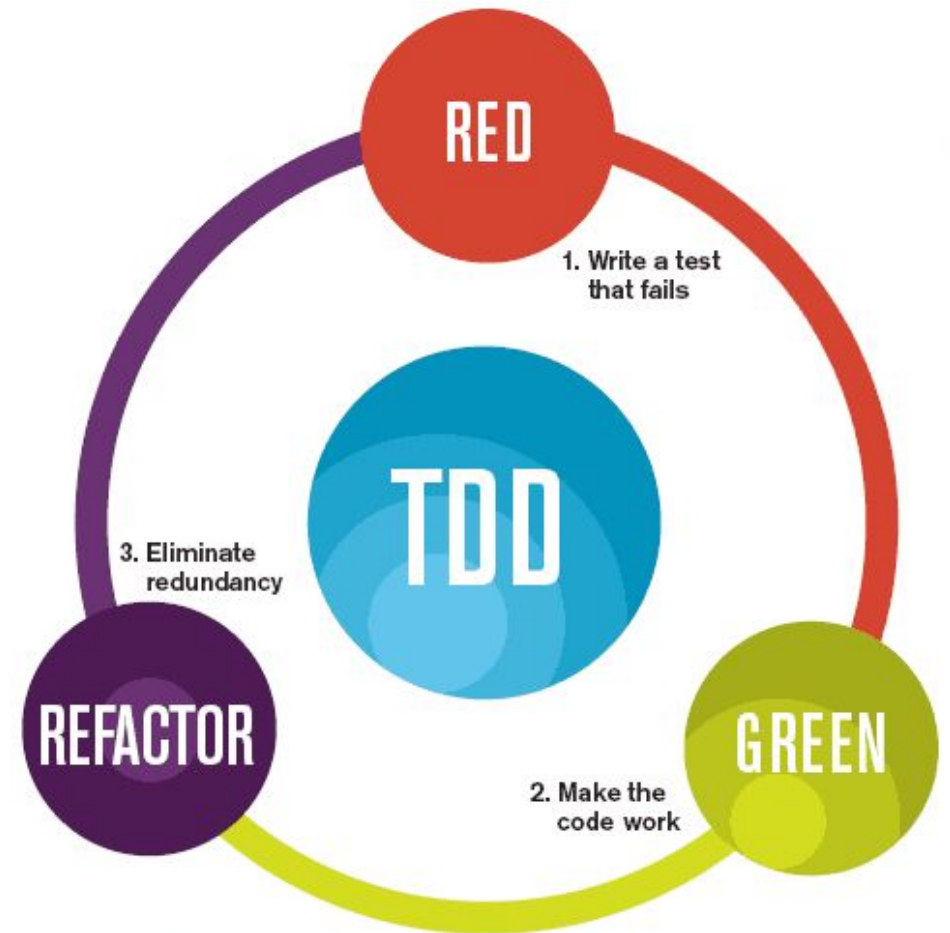
TDD - Test Driven Development

Agenda

- TDD - Some Background
- From Requirement to Technical Detail
- From Technical Detail to Tests
- Discuss Projects

Background

- Write tests first
- Make it pass by writing the minimum code
- Refactor if needed
- Repeat



The mantra of Test-Driven Development (TDD) is "red, green, refactor."

Why Do We Care?

- Many companies employ some sort of TDD
 - Often not as strict
- It makes you think of what you want to do before writing code
 - This is the most important skill that we want to impart to you

The Requirement

The user needs to be able to log in

What does this mean exactly?

- Does this imply that users can create accounts?
- How do we know that the user is logged in?
- Should users logout?

This is not necessarily the same for all applications

Let's make a list of all technical implications of this requirement

For Example

- Sending correct credentials will return an access token
 - Incorrect credentials will return an error
 - A different error when some of the credentials are missing?
- The access token enables access to protected route
 - Requests without a token or with an invalid token are blocked
- A request to log out with a valid token will disable the token

For Example

```
@pytest.fixture
def app():
    # Possible setup here
    app = create_app({'TESTING': True})
    yield app
    # Possible teardown here

@pytest.fixture
def client(app):
    return app.test_client()

@pytest.fixture
def test_user():
    return {
        'username': 'testuser',
        'password': 'testpass123'
    }
```

```
def test_login_success(client, test_user):
    """Test successful login"""
    # First register the user
    client.post('/api/auth/register', json=test_user)

    # Then try to login
    response = client.post('/api/auth/login',
                           json=test_user)
    )
    assert response.status_code == 200
    data = json.loads(response.data)
    assert 'access_token' in data
```

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Now let's implement tests for the rest of these